Image and Invertibility of n-tuple Vertex Graphs

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Abstract

Erdös, Alavi, Behzad, and Lick introduced the concept of a double vertex graph, which was then generalized to the n-tuple vertex graph by Wright as follows: For a graph G with vertex set $V(G)$ and edge set $E(G)$, its n-tuple $U_n(G)$ has vertex set $U_n(V(G))$ consisting of all n-element subsets of $V(G)$ and edge set $U_n(E(G))$ consisting of pairs of vertices matching all but one element where the distinct elements are adjacent vertices in $V(G)$.

In this paper I explore the n-tuple vertex graph as an operation over the space of connected, finite graphs (write $\mathbb{G}$), focusing on the questions:

(1) What is the image of U_n in $\mathbb{G}$?

(2) How can we define U_n^{-1}?

I establish some properties of graphs in the image of U_n and hence show a way to determine whether certain types of graphs are n-tuples of a graph $G \in \mathbb{G}$ for some n. I then develop a procedure to deconstruct n-tuples into their inverse image.

The n-tuple operation has applications in combinatorics and Ramsey theory, and is particularly useful in coloring problems.

This paper assumes rudimentary knowledge of graph theory and set theory.